DCM operation of Boost PFC Converter

Introduction

In the realm of power electronics, particularly for applications like charging infrastructure, Power Factor Correction (PFC) converters are crucial for ensuring that electrical loads draw current efficiently from the AC mains. Boost PFC converters are a common choice, often operating in Continuous Conduction Mode (CCM) to achieve high power factor and low harmonic distortion. However, CCM operation introduces significant control complexity, primarily due to the need for sophisticated control algorithms involving multiplier blocks and precise generation of unit sine waveforms. These requirements often necessitate high-end microcontrollers, increasing computational burden, memory usage, and overall system cost.

This lecture explores an alternative approach: the Discontinuous Conduction Mode (DCM) operation of Boost PFC converters. By intentionally allowing the inductor current to drop to zero within each switching cycle, DCM offers a pathway to simplify the control strategy. This simplification can lead to reduced computational demands and potentially lower cost implementations, making it an attractive option for certain applications despite some inherent trade-offs in performance. Understanding DCM operation is key to appreciating the design choices and compromises involved in practical PFC converter designs.

Core Concepts

Term	Definition	Significance
`DCM (Discontinuous Conduction Mode)`	An operating mode where the inductor current in a converter falls to zero during each switching cycle.	Simplifies control by eliminating the need for complex current loop feedback and multiplier blocks.
`PFC (Power Factor Correction)`	A technique used to improve the power factor of an AC load, making the current waveform more sinusoidal and in phase with the voltage.	Essential for meeting regulatory standards and ensuring efficient power delivery from the grid.

Term	Definition	Significance
`Multiplier Block`	A computational component in CCM PFC control loops that multiplies a voltage controller's output with a rectified sine wave.	A major source of computational complexity and cost in CCM PFC, which DCM aims to avoid.
`Unit Sine Waveform Generation`	The process of creating a precise sinusoidal reference signal, often synchronized with the input AC voltage.	Crucial for shaping the input current in CCM PFC, but challenging to implement accurately due to grid variations.
`THD (Total Harmonic Distortion)`	A measure of the harmonic content present in a waveform, indicating how much it deviates from a pure sine wave.	DCM operation typically results in higher THD compared to CCM due to the less perfectly sinusoidal current.
`Reverse Recovery Losses`	Energy losses in a diode when it switches from forward conduction to reverse blocking, due to stored charge.	Significantly reduced in DCM operation, especially if a diode is used as the output switch, as it turns off with zero current.
`Voltage Follower Mode`	An operating characteristic where the input current automatically follows the shape of the input AC voltage.	A key advantage of DCM, as it inherently achieves power factor correction without explicit current shaping control.
`RMS Current`	The root mean square value of current, representing its heating effect.	Higher in DCM components (inductor, switches) due to increased ripple, leading to higher conduction losses.

Detailed Analysis

1. Limitations of Continuous Conduction Mode (CCM) Boost PFC

Previous lectures covered the operation of boost PFC converters, including both diode-bridge-based and bridgeless topologies. In these CCM operations, the primary goal was to control the inductor current (`iL`) to ensure that the input current (`is`) drawn from the source was sinusoidal and in phase with the input voltage, thereby achieving unity power factor. This control typically involved a closed-loop system with a constant switching frequency. A key characteristic of CCM is that the inductor current never drops to zero, except possibly near zero-crossing points of the AC voltage.

However, achieving this in CCM comes with significant control complexity:
* **Multiplier Block Requirement**: The control loop in CCM often requires a "multiplier block." For a bridge boost PFC, this block multiplies the output of the voltage controller with the modulus of `sin(ωt)` . For a bridgeless PFC, it multiplies with `sin(ωt)` . This block is computationally intensive, demanding high-end microcontrollers or specialized analog components, which can increase cost and design complexity.
* **Unit Sine Waveform Generation**: Another challenge is the accurate generation of the "unit sine waveform" (`sin(ωt)` or `modulus of sin(ωt)`). This waveform must precisely track the actual input AC voltage, meaning there should be no phase angle difference. Grid voltages can vary in frequency and magnitude, making accurate generation difficult. It often requires sophisticated Phase-Locked Loops (PLLs) and filtering circuits, adding to the computational and hardware burden.
* **Computational Burden**: The combination of multiplier blocks and unit sine waveform generation leads to a substantial computational load, requiring significant memory space and processing power. This can limit the use of lower-cost microcontrollers.

To circumvent these complexities, an alternative approach, Discontinuous Conduction Mode (DCM), is explored, aiming to simplify the control structure.

2. DCM Operation Principle and Inductor Current Waveform

In DCM, the fundamental difference from CCM is that the inductor current (`iL`) is allowed to fall to zero within every switching cycle. This characteristic inherently simplifies the control strategy.

Consider a single switching cycle (`Ts`). The inductor current waveform in DCM typically exhibits the following behavior:
* **Switch S1 ON (TS1 on)**: When switch S1 is turned on, the input voltage (`vs` or

`vd = |vs|`) is applied across the inductor. The inductor current starts from zero and ramps up linearly with a positive slope, reaching a peak value (`Ip`).
* **Switch S1 OFF, S2 ON (TS2 on)**: After `TS1 on` , S1 is turned off, and S2 (or the diode in its place) turns on. The voltage across the inductor becomes `vs - Vo` (or `|vs| - Vo`). Since the output voltage `Vo` is typically greater than the peak input voltage `vs,pk` , a negative voltage is applied across the inductor. The inductor current then ramps down linearly with a negative slope.
* **Current Reaches Zero**: Crucially, in DCM, the inductor current reaches zero before the end of the switching cycle.
* **Zero Current Period**: After the current reaches zero, it remains at zero until the next switching cycle begins, when S1 is turned on again. During this period, no current is drawn from the source.

This "discontinuous" nature of the inductor current is the defining feature of DCM.

3. Inductor Current Behavior and Voltage Following

Let's analyze the inductor current behavior more formally.
During the `TS1 on` period, when S1 is on, the voltage across the inductor (`vL`) is `vd` , which is `|vs|` .
The change in inductor current (`di`) is from 0 to `Ip` .
Using the inductor voltage-current relationship `vL = L * di/dt` :
`|vs| = L * (Ip / TS1 on)`
From this, the peak inductor current `Ip` can be expressed as:
`Ip = (|vs| * TS1 on) / L`

If the `TS1 on` duration is kept constant, then the peak inductor current (`Ip`) directly follows the modulus of the input AC voltage (`|vs|`). This means that if the input AC voltage is sinusoidal, the peak of the inductor current will also vary sinusoidally. This is a critical observation because it implies that the input current's peak is inherently shaped to follow the AC voltage, moving towards a voltage follower mode.

4. Average Inductor Current and Input Current Characteristics

To understand the overall input current behavior, we need to calculate the average inductor current (`iL(t) average`) over a switching cycle (`Ts`). The average current is derived from the triangular waveform of the inductor current.

The average value of the inductor current over `Ts` can be approximated as:
`iL(t) average = (1/2) * Ip * (D1 * Ts + D2 * Ts) / Ts`
where `D1 = TS1 on / Ts` and `D2 = TS2 on / Ts` are the duty cycles for the current rise and fall phases, respectively.

Applying volt-second balance across the inductor over a switching cycle (the net change in inductor flux is zero over a period):
`|vs(t)| * D1 * Ts + (-Vo + |vs(t)|) * D2 * Ts = 0`
From this, `D2` can be expressed as:
`D2 = (|vs(t)| / (Vo - |vs(t)|)) * D1`

Substituting `D2` back into the average current equation and simplifying, the average inductor current `iL(t) average` over the `Ts` duration is found to be proportional to `|vs(t)|` and `D1^2`, and inversely proportional to `L`. More specifically, the lecture states the average input current `is(t)` (which is `iL(t)` when `vs > 0` and `-iL(t)` when `vs < 0`) is:
`is(t) average = (1/2) * vs,pk * sin(ωt) * D1^2 * Ts * (Vo / (Vo - |vs,pk * sin(ωt)|)) / L` (with `D1` and `Ts` constant).

This complex expression indicates that while the peak inductor current `Ip` varies sinusoidally, the average input current `is(t)` is not perfectly sinusoidal but is "close to sinusoidal." This means it will have a sinusoidal behavior but with some distortion, leading to a higher Total Harmonic Distortion (THD) compared to CCM.

5. Advantages of DCM Operation

The DCM operation offers several compelling advantages, primarily stemming from its simplified control:
* **Automatic Current Following**: The input current drawn from the source automatically follows the shape of the AC input voltage. This inherent voltage-follower mode eliminates the need for complex current shaping control loops.
* **Elimination of Multiplier Block**: The most significant advantage is the removal of the multiplier block from the control scheme. This drastically reduces computational requirements and simplifies the controller design.
* **No Unit Sine Waveform Generation**: There is no need to generate a precise unit sine waveform (`sin(ωt)`). This avoids the challenges associated with tracking grid variations and implementing complex PLLs and filters.
* **Simplified Closed-Loop Control**: The overall closed-loop control becomes much simpler. Often, only a single voltage control loop is needed to regulate the output voltage, with the input current shaping happening automatically.
* **Reduced Switching Losses**:
* When switch S1 turns on, it does so with zero current (Zero Current Switching - ZCS), as the inductor current has already dropped to zero.
* If a diode is used in place of S2, its reverse recovery losses are nearly zero because it turns off when the current through it is zero. This is a significant benefit for high-frequency applications.

6. Disadvantages of DCM Operation

Despite its advantages, DCM operation comes with certain drawbacks:
* **Higher THD**: The average input current is only "close to sinusoidal" and not perfectly sinusoidal. This results in a higher Total Harmonic Distortion (THD) in the input current compared to CCM, which might not meet stringent harmonic standards without additional filtering.
* **Increased Ripple**: The inductor current in DCM has a larger peak-to-peak ripple compared to CCM, as it swings from zero to a peak and back to zero in every cycle.
* **Higher RMS Currents**: Due to the increased ripple and the discontinuous nature, the RMS currents flowing through the inductor, switches (S1, S2), and diodes are higher than in CCM. Higher RMS currents lead to increased conduction losses and require components with higher current ratings, potentially increasing their size and cost.
* **Higher Switching Losses for S1 Turn-Off**: While S1 turns on with zero current, it turns off with the peak inductor current (`Ip`). This can lead to higher turn-off switching losses for S1, and careful gate driver design is needed to manage parasitic effects.
* **Larger Inductor Size (for same ripple)**: To limit the peak current and ripple to acceptable levels, a larger inductance value might be required compared to CCM for a given power level, potentially increasing the physical size of the inductor.

Key Takeaways

- [ ] DCM operation allows the inductor current to drop to zero in every switching cycle, simplifying control.
- [ ] CCM PFC requires complex multiplier blocks and accurate unit sine waveform generation, leading to high computational burden.
- [ ] In DCM, the peak inductor current (`Ip`) inherently follows the modulus of the input AC voltage, enabling a voltage-follower mode.
- [ ] The average input current in DCM is "close to sinusoidal" but not perfectly so, resulting in higher THD compared to CCM.
- [ ] DCM eliminates the need for multiplier blocks and unit sine waveform generation, simplifying the control circuit and reducing computational demands.
- [ ] DCM operation significantly reduces switching losses, especially reverse recovery losses in diodes, as components turn off at zero current.
- [ ] Disadvantages of DCM include higher THD, increased current ripple, and higher RMS currents in components, leading to higher conduction losses.

- [ ] The simplified control of DCM often requires only a single voltage control loop, making it suitable for low-cost microcontroller implementations.

Conclusion

The Discontinuous Conduction Mode (DCM) operation of Boost PFC converters presents a compelling trade-off between control complexity and performance. By allowing the inductor current to reset to zero in each switching cycle, DCM inherently simplifies the control strategy, eliminating the need for computationally intensive multiplier blocks and the challenging generation of precise unit sine waveforms. This simplification translates into lower computational requirements, potentially enabling the use of more cost-effective microcontrollers and reducing overall system design complexity. Furthermore, DCM offers advantages such as reduced switching losses, particularly the elimination of reverse recovery losses in diodes, which can enhance efficiency and reliability.

However, these benefits come with inherent compromises. The input current in DCM, while following the general sinusoidal shape of the AC voltage, is not perfectly sinusoidal, leading to higher Total Harmonic Distortion (THD). Additionally, the increased current ripple results in higher RMS currents through the converter components, which can lead to greater conduction losses and necessitate larger component ratings. Therefore, the choice between CCM and DCM for a Boost PFC converter depends heavily on the specific application requirements, balancing the need for high power quality and efficiency against the desire for simplified control and reduced cost. For applications where strict THD requirements are relaxed or where cost and control simplicity are paramount, DCM offers a viable and attractive solution in charging infrastructure and other power conversion systems.